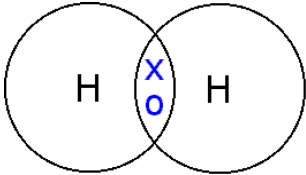


| Question |     |       |    | Marking details                                                                                                                                                                     | Marks Available |          |          |           |          |      |
|----------|-----|-------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|------|
|          |     |       |    |                                                                                                                                                                                     | AO1             | AO2      | AO3      | Total     | Maths    | Prac |
| 5 FT     | (a) |       |    |  <p>Accept anywhere in the overlap region including on the line<br/>Could be two dots or two x</p> | 1               |          |          |           |          |      |
|          | (b) | (i)   |    | $436 + 242 (1) = 678 (1)$<br>Accept answers of 436 or 242 for (1) only                                                                                                              |                 | 2        |          |           | 2        |      |
|          |     | (ii)  |    | $431 + 431 (1) = 862 (1)$<br>Accept answer of 431 for (1) only                                                                                                                      |                 | 2        |          |           | 2        |      |
|          |     | (iii) | I  | A                                                                                                                                                                                   |                 |          | 1        |           |          |      |
|          |     |       | II | More energy given out than taken in /Exothermic (1)<br>So the line ends up lower than it started / -184 (1)                                                                         |                 | 2        |          |           |          |      |
|          | (c) |       |    | More taken in decrease thermal runaway                                                                                                                                              | 4               |          |          |           |          |      |
|          |     |       |    | <b>Question 5 total</b>                                                                                                                                                             | <b>5</b>        | <b>6</b> | <b>1</b> | <b>12</b> | <b>4</b> |      |